

Jonathan Ty

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EDUCATION

University of California, San Diego

La Jolla, CA

B.S. in Computer Science, Minors in Cognitive Science and Business Economics

Sept 2023 – June 2027

- GPA: 3.526 Cumulative | 3.515 Major
- Honors: Regent's Scholar, Alan Turing Memorial Scholar, TAPIA 2024 Grant | Cert: AWS Cloud Foundations

EXPERIENCE

Software Development Intern

June 2025 – Sept 2025

Aimia, AI-driven career development platform

San Diego, CA

- Built user-facing features and authentication flows end-to-end in a production Next.js/React application, shipping responsive UI and session management from design through deployment
- Designed and integrated RESTful API services with Express.js and MongoDB, implementing JWT-secured CRUD endpoints and managing application state with custom React hooks across 15+ modules
- Collaborated in an agile team using Git-based workflows, code review, and iterative feature delivery

Software Engineer Intern

June 2026 – Present

San Diego Supercomputer Center (SDSC)

San Diego, CA

- Selected for a team-based agile software development internship building and maintaining internal research data services and tooling
- Contributing to full-stack application development and data infrastructure in a collaborative engineering environment

Instructional Assistant: Data Structures & Algorithms

January 2026 – Present

UC San Diego, Computer Science & Engineering

La Jolla, CA

- Instructional Assistant for Professor Paul Cao across three consecutive terms, supporting 400+ students in Data Structures, Algorithms, and object-oriented design in Java
- Debug student code and give targeted feedback on algorithmic complexity, data structure design, and OOP patterns; graded 500+ exams and assignments

Undergraduate Research Assistant — Software & Data Pipelines

June 2026 – Present

UC San Diego, Cognitive Science Department – Chiba Lab

La Jolla, CA

- Core software developer building Python signal-processing pipelines for time-series physiological data in a longitudinal child-development study
- Engineer data synchronization tooling integrating multimodal datasets (video, behavioral annotations, wearable biosensor signals), with preprocessing, quality control, and documentation

Undergraduate Research Assistant — Human Activity Recognition

Sept 2024 – May 2025

UC San Diego, Computer Science & Engineering (ERSP)

La Jolla, CA

- Built Python data pipelines to parse, align, and validate 9-axis IMU sensor data across multiple labeling methods; trained and evaluated LSTM models reaching 81% accuracy
- Designed an audio-based labeling system that cut data-annotation time by over 70% through automated signal alignment; co-authored a resulting paper submitted to CHI and IMWUT

PROJECTS

Cookie Jar | JavaScript, Chrome MV3, Canvas API

Feb 2026

- Built a Chrome Manifest V3 extension that classifies and blocks cookies in real time, suppresses consent banners, and renders a Canvas API stats dashboard — won 2nd place at Design Frontiers 2026 (Deco Designathon at UCSD)
- Implemented across content scripts, background service workers, and popup UI; currently awaiting Chrome Web Store approval

CertConvert | Claude API, TypeScript, Vercel, Streaming, Multi-Agent

2026

- Built a dual-agent document conversion system at the Claude Builders Club Hackathon, configuring Claude Opus and Haiku agents via YAML with distinct roles in a managed-agents pipeline
- Implemented streaming API routes for real-time output delivery and deployed the full application to Vercel end-to-end

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL, HTML/CSS

Web & Full-Stack: React, Next.js, Node.js, Express.js, FastAPI, MongoDB, REST APIs, JWT Auth

Tools & Platforms: Git, Docker, AWS, Linux, Vercel, Jira, Confluence, Figma

ML & Data: PyTorch, Scikit-Learn, Pandas, NumPy, Claude API, LSTMs

Relevant Coursework: *Data Structures & Algorithms, Software Engineering, Operating Systems, Machine Learning (CSE 151A), Recommender Systems & Web Mining (CSE 158), Data Science in Practice, Cryptography*